G_{Ep}/G_{Mp} with an 11 GeV Electron Beam in Hall C

E.J. Brash, M.K. Jones,
C.F. Perdrisat, V. Punjabi

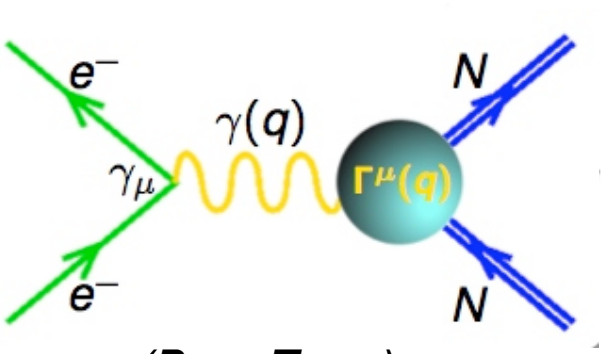
and the Gep-IV Collaboration

Based on LoI 12-06-103 to PAC30

Elastic EM Form Factors

- Fundamental to our understanding of nucleon structure
  - At low Q^2 (larger distance), they shed information on the pion cloud.
  - At intermediate/large Q^2 (shorter distance), they contain information on the quark and gluon structure
- At JLab energies, we gain knowledge about the non-perturbative structure of the nucleon
- Together with other observables (in DIS, SIDIS, DVCS, RCS, etc.), we develop a greater understanding of the nucleon's rich internal structure

Elastic Form Factors



(Born Term)

Nucleon current operator (Dirac & Pauli)

$$\Gamma^{\mu}(q) = \gamma^{\mu} F_1(q^2) + \frac{i}{2M_N}\sigma^{\mu\nu} q_\nu F_2(q^2)$$

$F_1(Q^2)$: non spin-flip Dirac form factor

$F_2(Q^2)$: spin-flip Pauli form factor

$$F_1^p(0) = 1, \quad F_1^n(0) = 0$$
$$F_2^p(0) = \kappa_p, \quad F_2^n(0) = \kappa_n$$

Electric form factor: $G_E(Q^2) = F_1(Q^2) - \tau F_2(Q^2)$

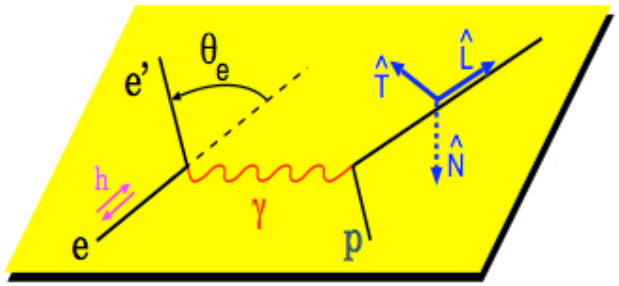
Magnetic form factor: $G_M(Q^2) = F_1(Q^2) + F_2(Q^2)$

where $\tau = \frac{Q^2}{4M_p^2}$

- Details of nucleon substructure are in the Q^2 evolution of $F_1(Q^2)$ and $F_2(Q^2)$.

Recoil Polarization Method

- Elastic $\vec{e}p \rightarrow e\vec{p}$ (A. I. Akhiezer and M. P. Rekalo, Sov. J. Part. Nuc. **3**, (1974) 277; and Arnold, Carlson and Gross, Phys. Rev. **C23** (1981) 363):



$$P_n = 0$$

$$I_0 P_t = -2\sqrt{\tau(1+\tau)}\, G_{Ep} G_{Mp} \tan\tfrac{\theta_e}{2}$$

$$I_0 P_\ell = \tfrac{1}{M_p}(E_e + E_{e'})\sqrt{\tau(1+\tau)}\, G_{Mp}^2 \tan^2\tfrac{\theta_e}{2}$$

$$I_0 = G_{Ep}^2 + \tau G_{Mp}^2 \left[1 + 2(1+\tau)\tan^2\tfrac{\theta_e}{2}\right]$$

Relative Measurement:

$$\frac{G_{Ep}}{G_{Mp}} = -\frac{P_t}{P_\ell}\frac{(E_e + E_{e'})}{2M_p}\tan\left(\frac{\theta_e}{2}\right)$$

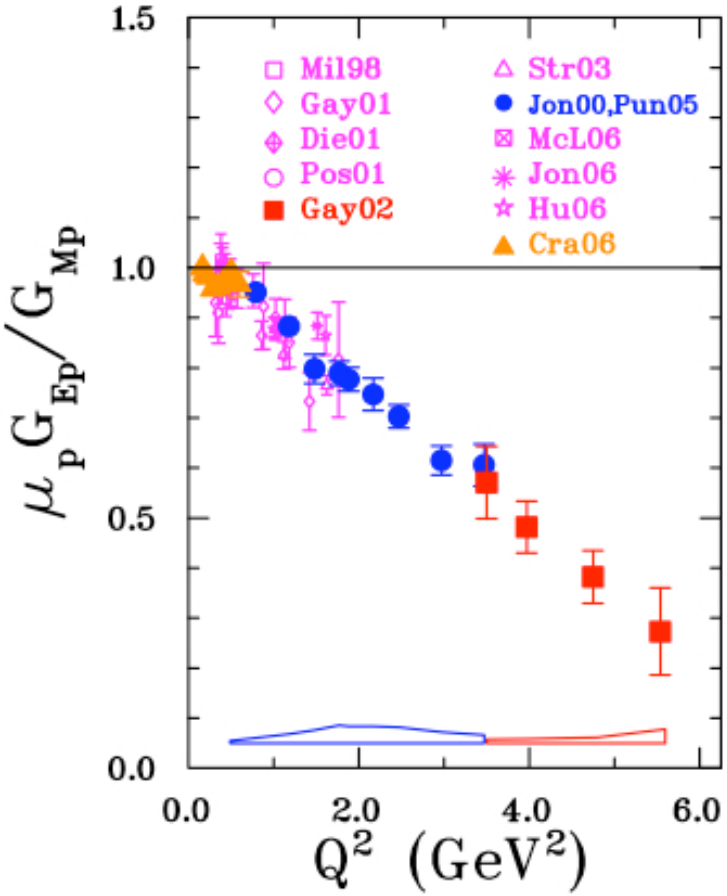
Recoil Polarization Measurements

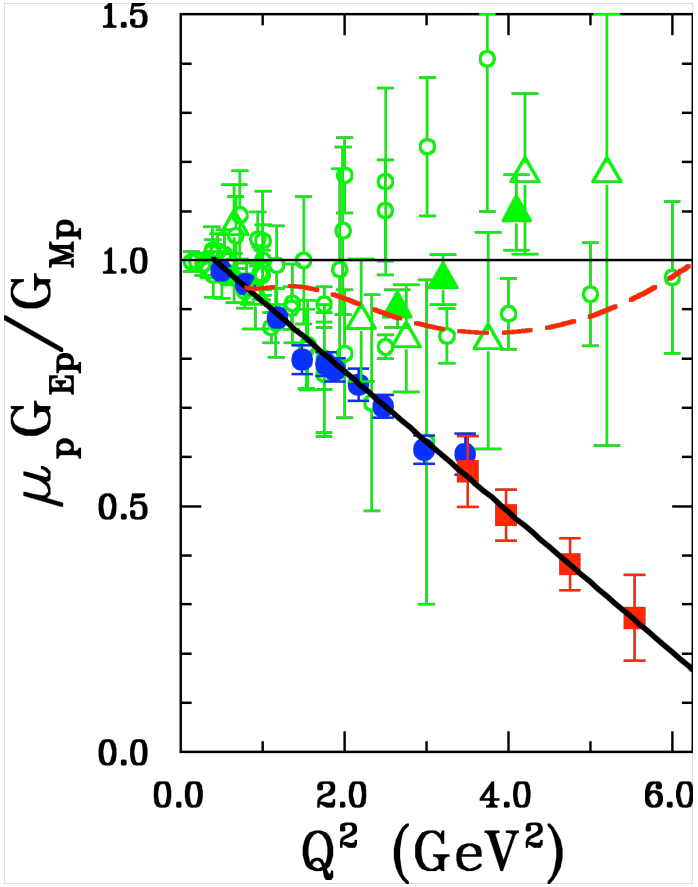
Famous Papers (250-499 citations)

M.K. Jones et al., Phys.Rev.Lett., 84 (2000) 1398 (G_{Ep}-I)

O. Gayou et al., Phys.Rev.Lett., 88 (2002) 092301. (G_{Ep}-II)

(829 total citations, 2 Ph.D.'s, 3 Masters, ~8 UG theses, several tenure track faculty and staff appointments)

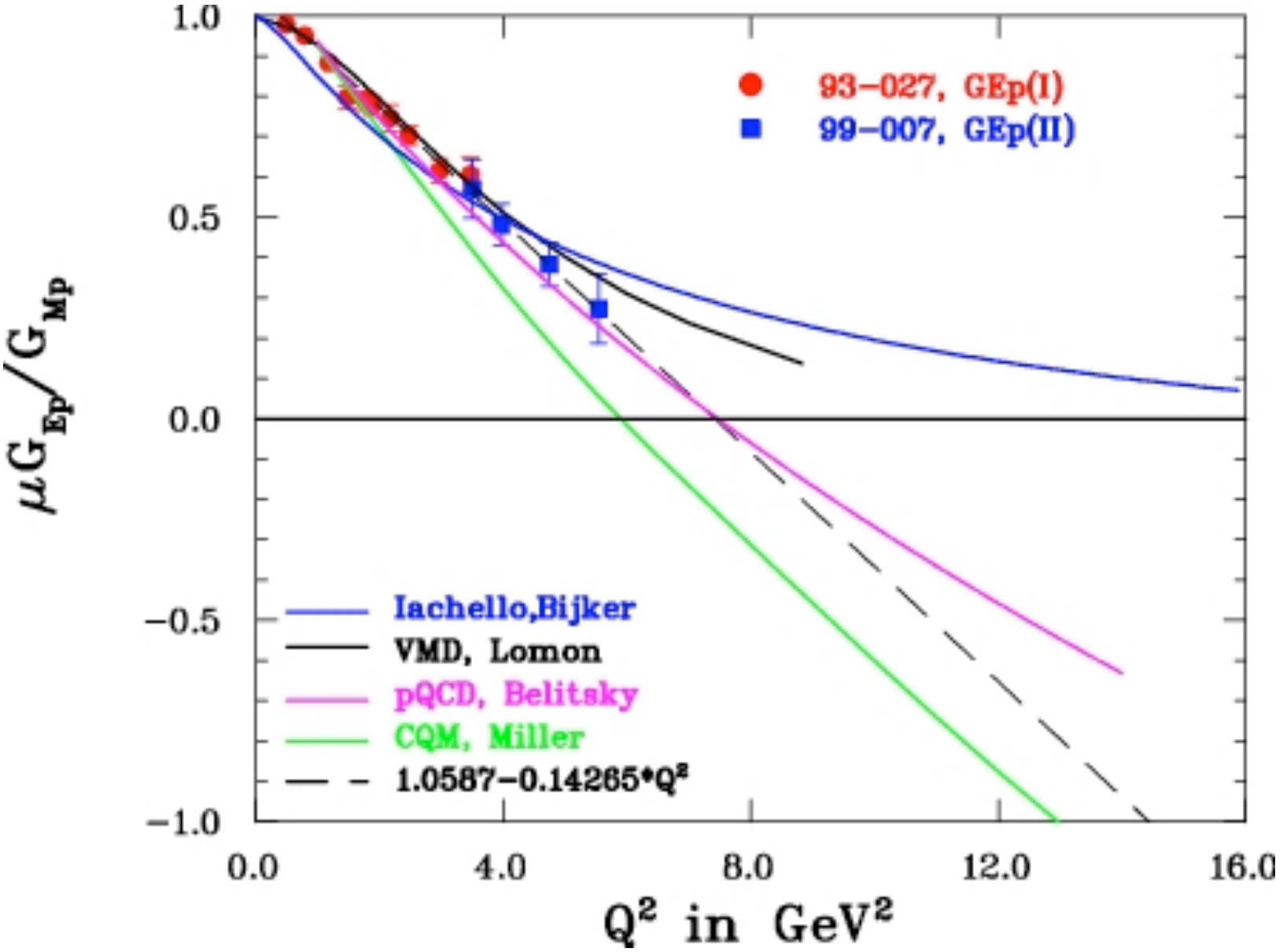




Theoretical Progress

- VMD-based models
  - Describe all four nucleon FF's well
  - Tend to favour ratio reaching a constant value at intermediate Q^2
- rCQM
  - Show the importance of relativistic dymamics
- pQCD-inspired models
  - Predict logarithmic scaling behaviour of F_2/F_1 at intermediate Q^2 (Belitsky and Ji) -> related to quark OAM
- GPD-inspired models
  - Show a connection with OAM of the quarks in the nucleon
  - FF's provide important constraints on GPD's
  - Behaviour of G_{Ep}/G_{Mp} at intermediate Q^2 related to u/d ratio at small distances (Miller)
- Lattice QCD Models
  - Good progress already, and will get much better in the future

Recoil Polarization Measurements



G_{Ep}-III Experiment at JLab

The ratio G_{Ep}/G_{Mp} was measured with the recoil polarization technique at Q^2 of 5.2, 6.8 and 8.54 GeV^2 in Hall C at JLab, between October 2007 and June 2008. Also G_{Ep}/G_{Mp} at a constant Q^2 of 2.49 GeV^2 for 3 kinematics (3 values of ε).

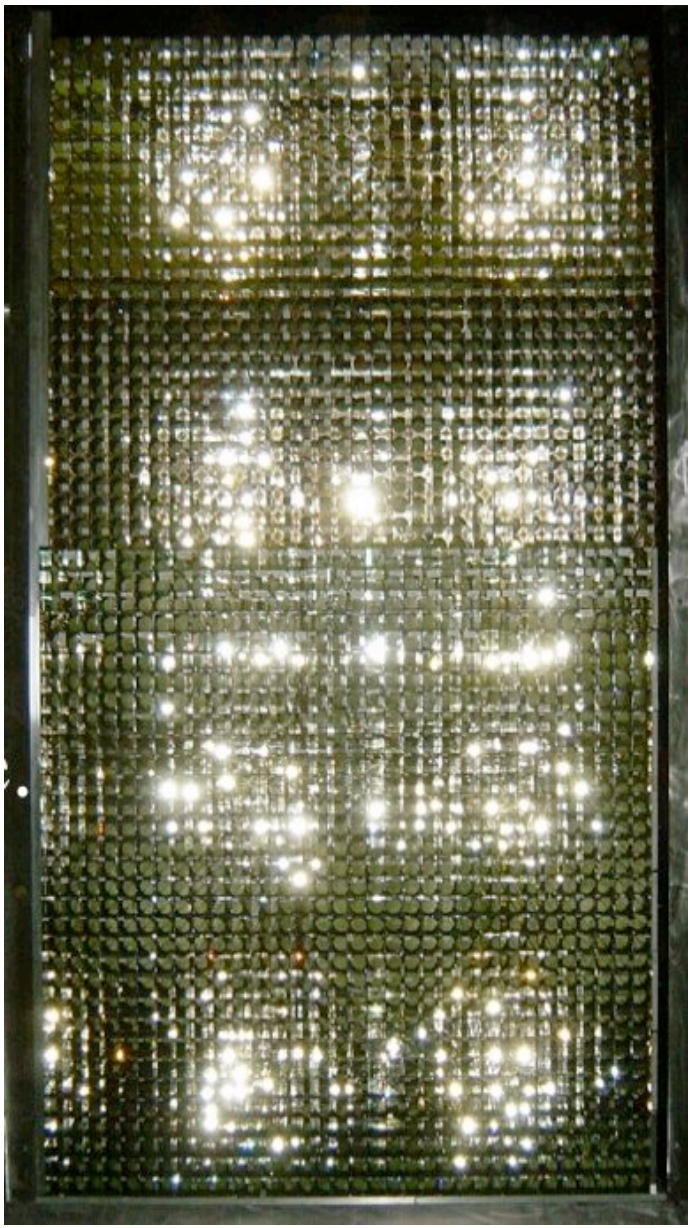
These experiments used the High Momentum Spectrometer (HMS) to detect proton; a new double focal plane polarimeter (FPP) in the focal plane of the HMS measured the polarization of the recoil proton.

A large area Electromagnetic Calorimeter (BigCal) assembled for the experiment was used to detect the elastically scattered electrons in coincidence with the protons.

BigCal in Hall C



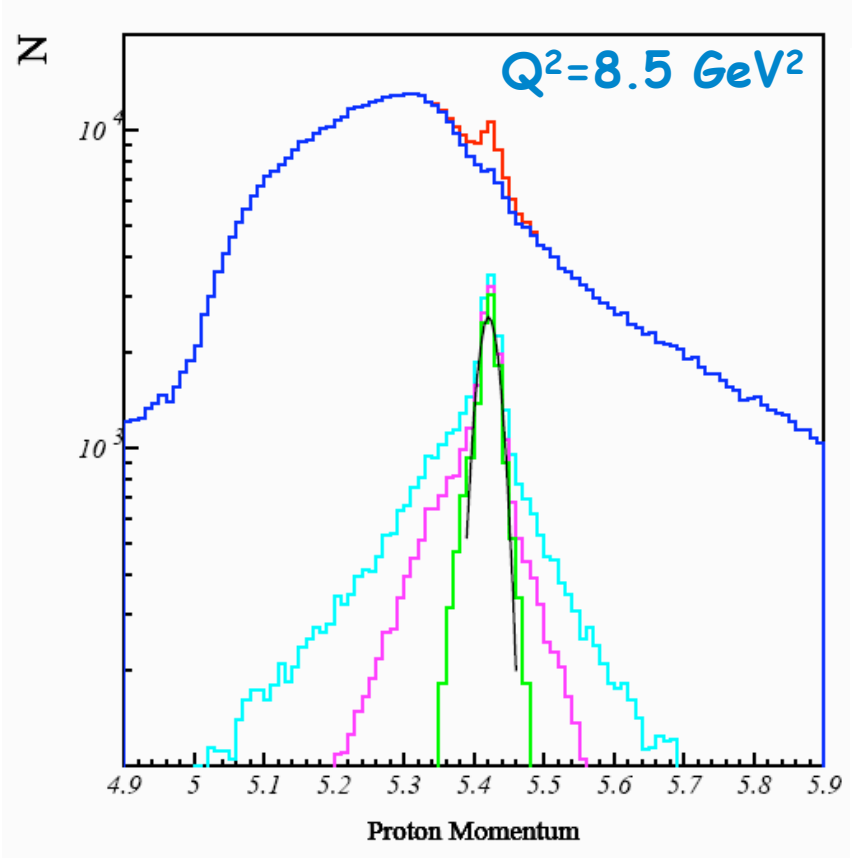
BigCal glass



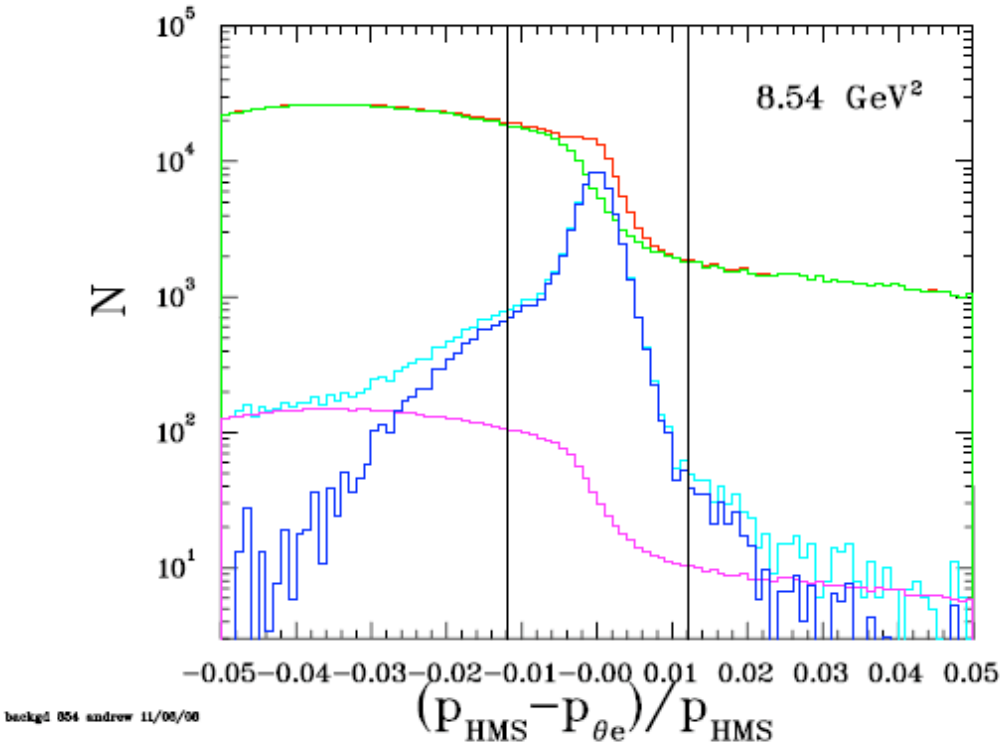
Double Focal Plane Polarimeter



Identification of Elastic Events: Proton Momentum Spectrum

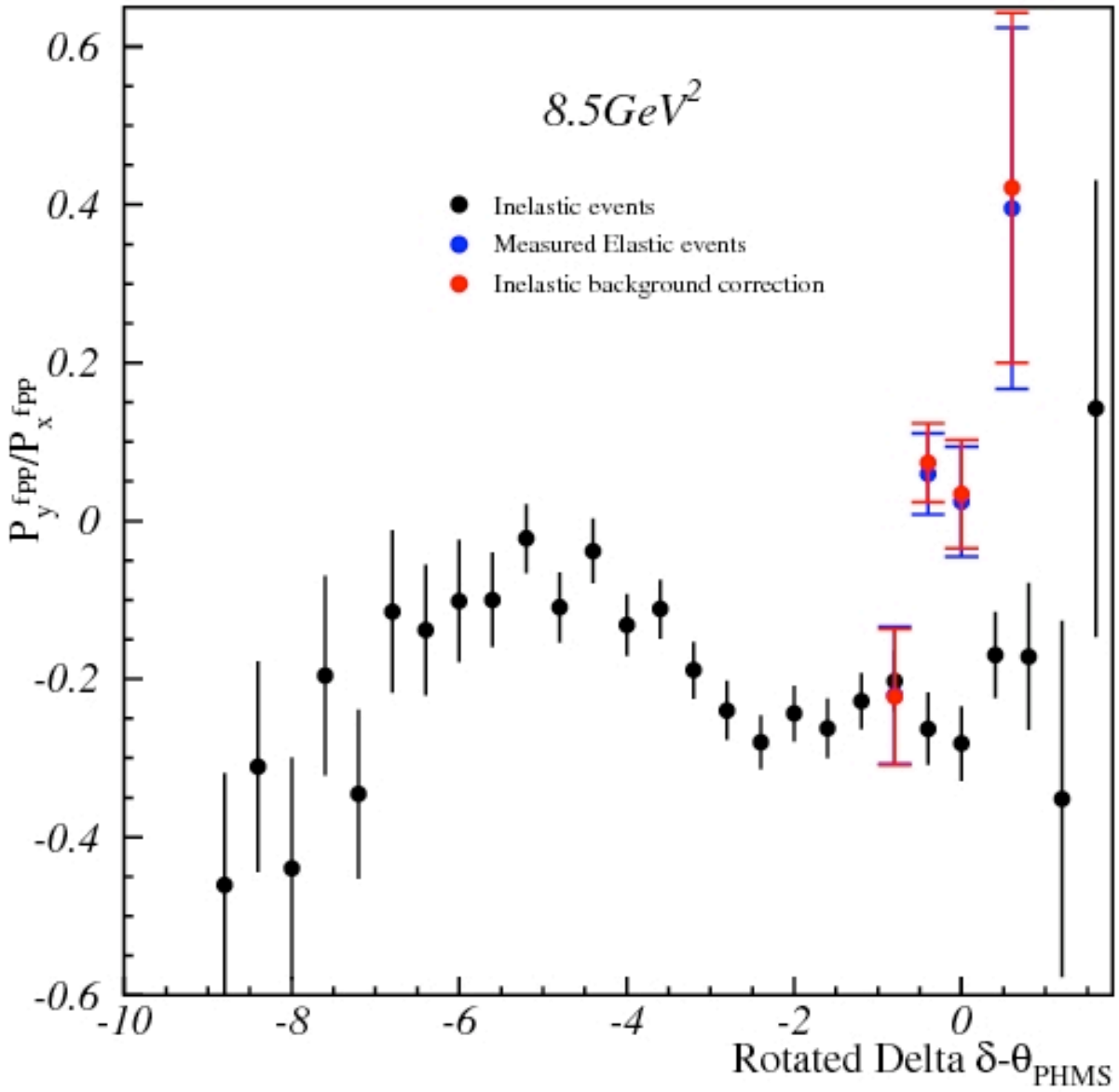


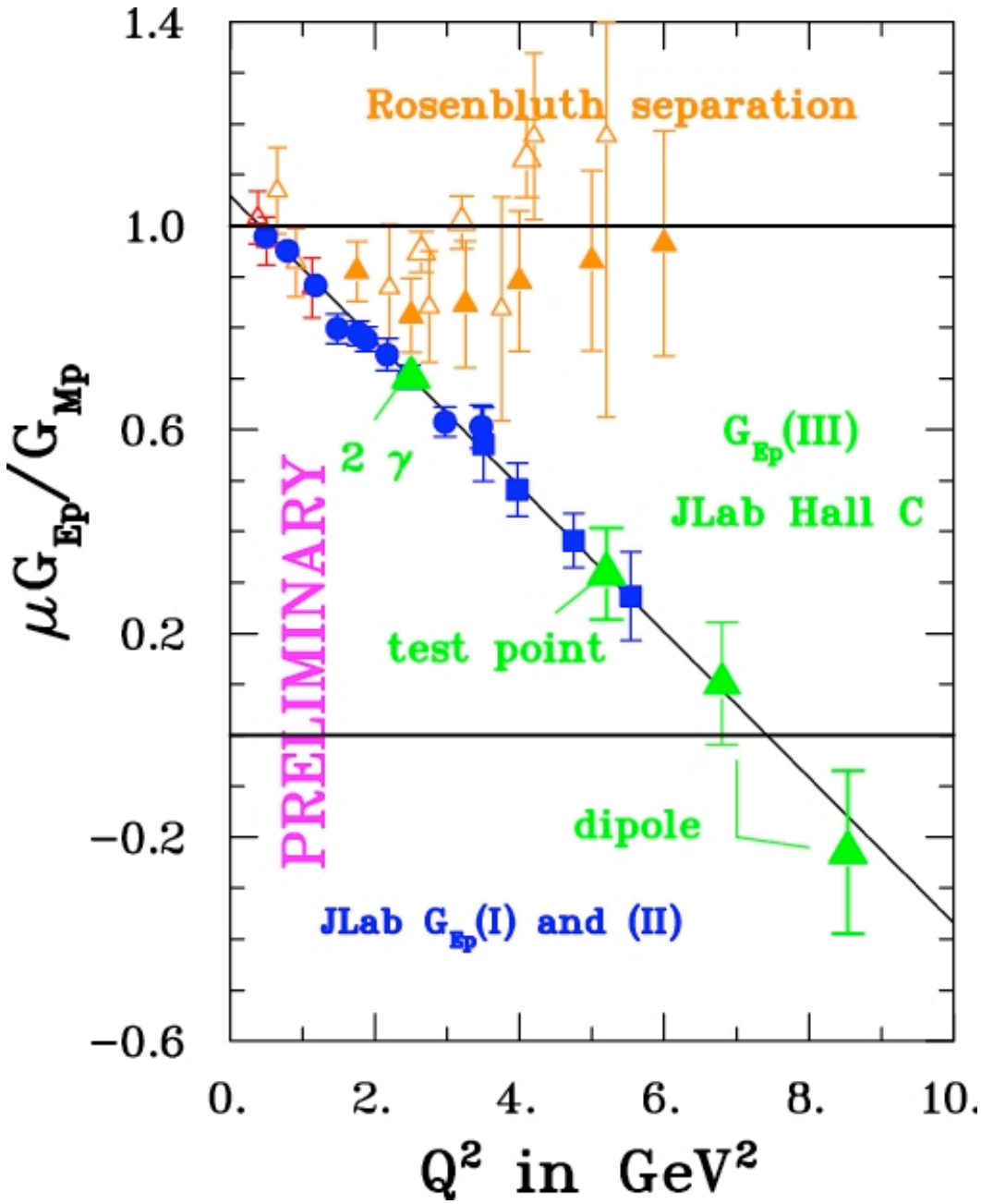
Red : all events,
Cyan: with $\delta - \theta$ cut,
Magenta: requiring co-planarity,
Green: loc. in BigCal and polar angle correlation
Blue: Non-elastic



Red : all events,
Cyan: x-y localization in BigCal
Blue: with Background subtraction

Magenta: Background Estimate … ~1%


8.5GeV²
Inelastic events
Measured Elastic events
Inelastic background correction
P_y^fpp/P_x^fpp
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-10
-8
-6
-4
-2
0
Rotated Delta δ-θ_PHMS

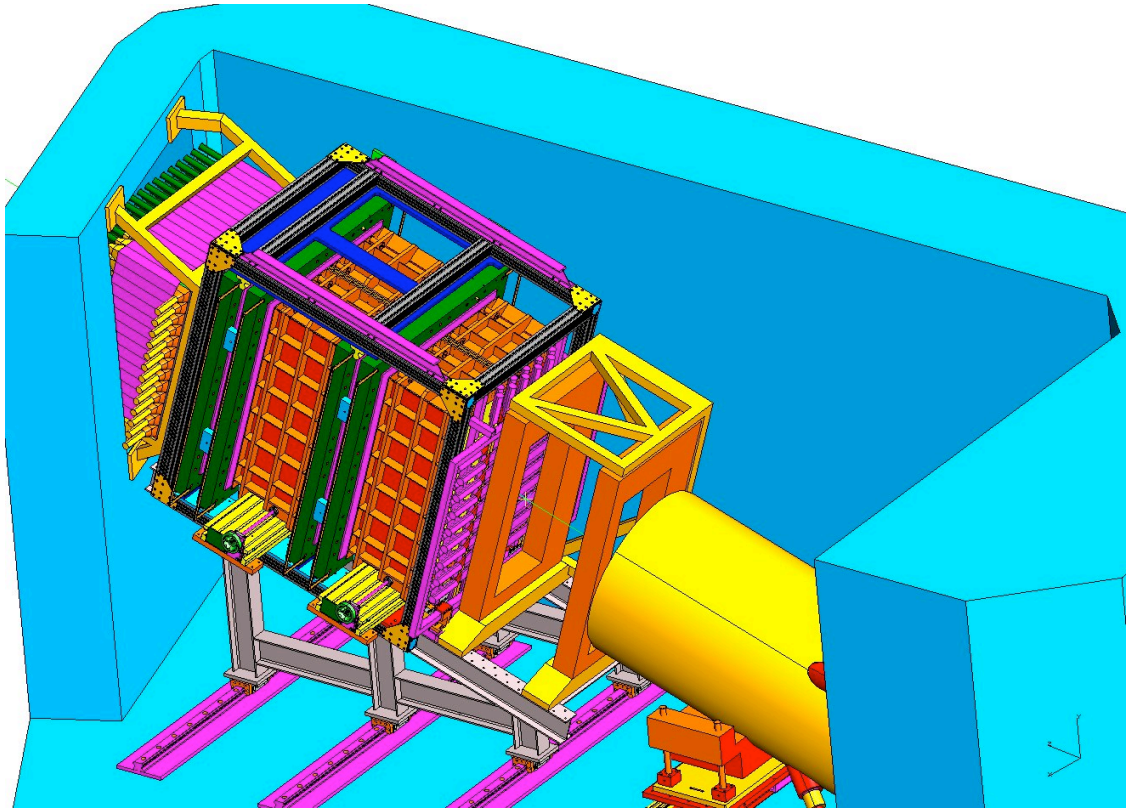

1.4
1.0
0.6
0.2
−0.2
−0.6
0.
2.
4.
6.
8.
10.
$\mu G_{Ep}/G_{Mp}$
Q^2 in GeV^2
Rosenbluth separation
2 γ
G_{Ep}(III)
JLab Hall C
test point
dipole
PRELIMINARY
JLab G_{Ep}(I) and (II)

Gep-IV: Basic Plan

- Extend the measurement of G_{Ep}/G_{Mp} to the largest value of Q^2 possible using base equipment together with existing dectectors in Hall C

- Hall C SHMS, equipped with the existing Hall C FPP, for proton detection

-The existing BigCal detector for electron detector - perfect match to the SHMS



Gep-IV: Issues for Consideration

1. Spin Precession in the SHMS

2. Radiation Damage to BigCal

3. CH_2 Analyzing Power

Gep-IV: Spin Precession in the SHMS

- The proton spin components precess in the magnetic elements of the SHMS, so that:

$$\begin{pmatrix} \mathbf{P_n^{fp}} \\ \mathbf{P_t^{fp}} \\ P_l^{fp} \end{pmatrix} = \begin{pmatrix} S_{nn} & \mathbf{S_{nt}} & \mathbf{S_{nl}} \\ S_{tn} & \mathbf{S_{tt}} & \mathbf{S_{tl}} \\ S_{ln} & S_{lt} & S_{ll} \end{pmatrix} \begin{pmatrix} P_n^{tar} \\ \mathbf{P_t^{tar}} \\ \mathbf{P_l^{tar}} \end{pmatrix}$$

- Horizontal bender leads to mixing of long. and transverse components

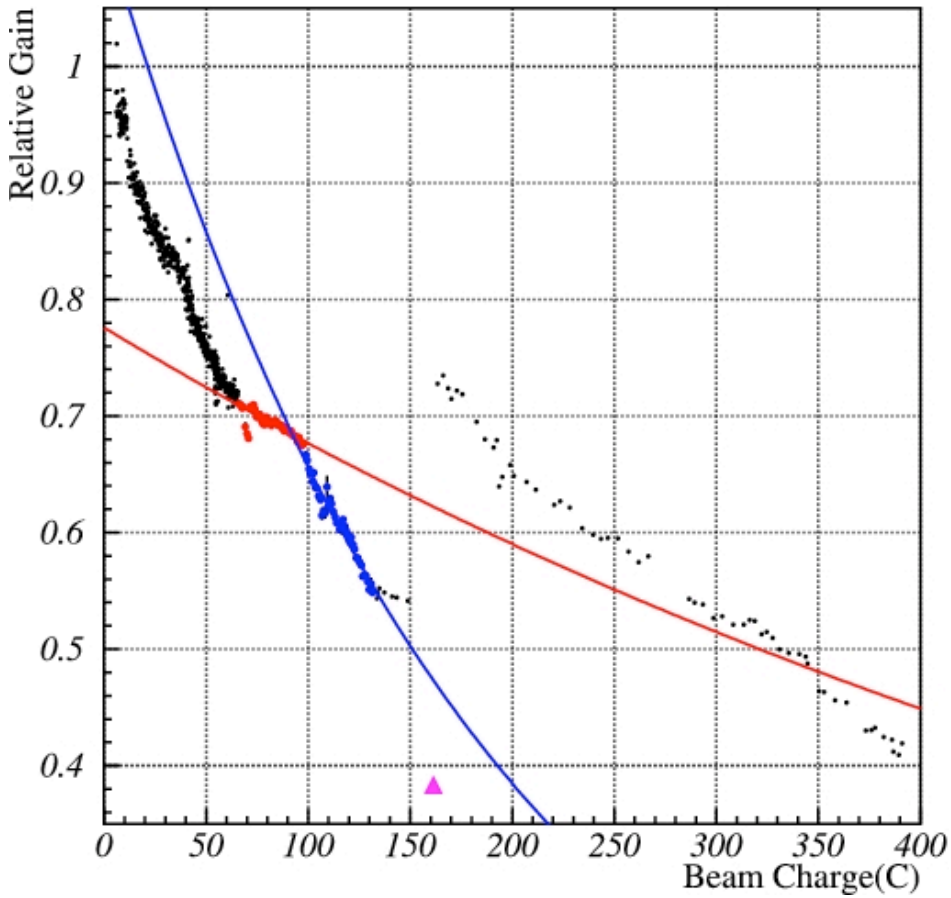
$$(\Delta P_t^{tar})^2 = \frac{2}{NA^2}(\cos^2\phi_{hb} + \frac{\sin^2\phi_{hb}}{\sin^2\phi_d})$$

$$(\Delta P_l^{tar})^2 = \frac{2}{NA^2}(\sin^2\phi_{hb} + \frac{\cos^2\phi_{hb}}{\sin^2\phi_d})$$

For $Q^2 = 13$ GeV^2, $\sin\phi_d \approx 1$, and thus $\cos^2\phi_{hb} + \frac{\sin^2\phi_{hb}}{\sin^2\phi_d} \approx 1$

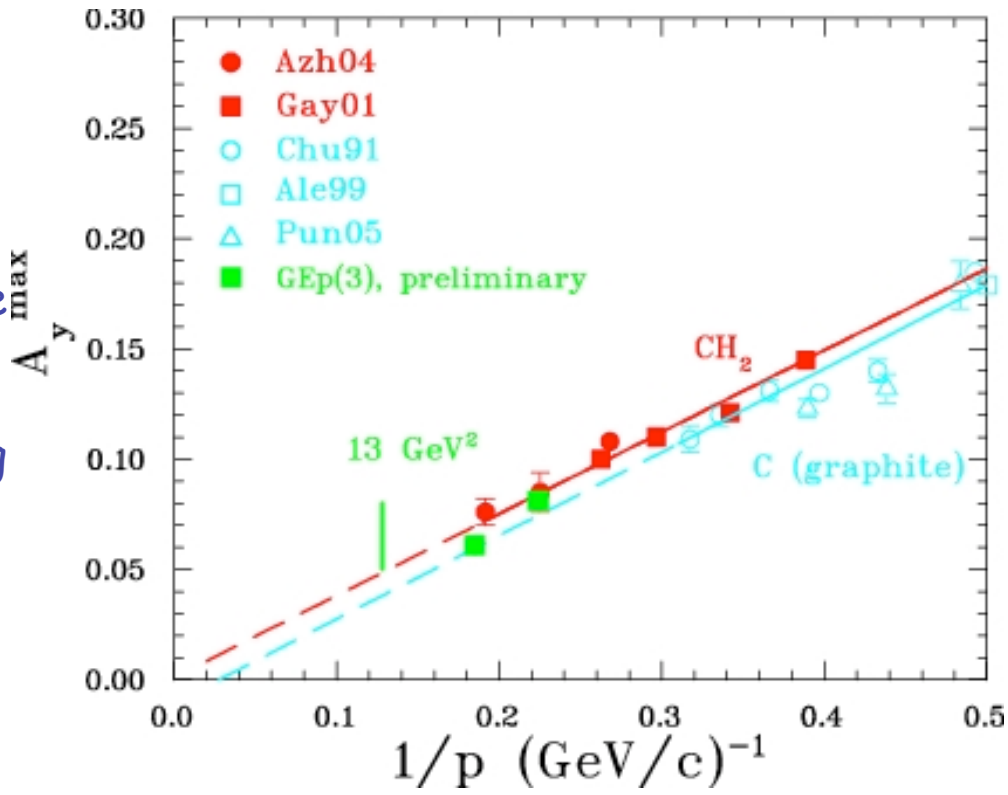
Gep-IV:BigCal Radiation Damage

- Affects energy resolution
  - we are fairly insensitive to this.
- Main concern:
  - relatively high hardware threshold to keep the BigCal rates low
- Result of GEANT Simulation:
  - Curing about once per week in Gep-IV
- Use maintenance days -
  - need four hours of curing to recover one week of damage



Gep-IV: CH_2 Analyzing Power

- As a by-product of the polarization transfer experiments, we can extract the (average/maximum) CH_2 analyzing power in the FPP
- Empirically, the maximum analyzing power scales as 1/p, the shape of the distribution scales in a similar manner; this allows us to make accurate predictions of the analyzing power at various momenta
- In addition, full GEANT3 simulations have been performed to estimate the scattering efficiency (describes Gep-III data well)



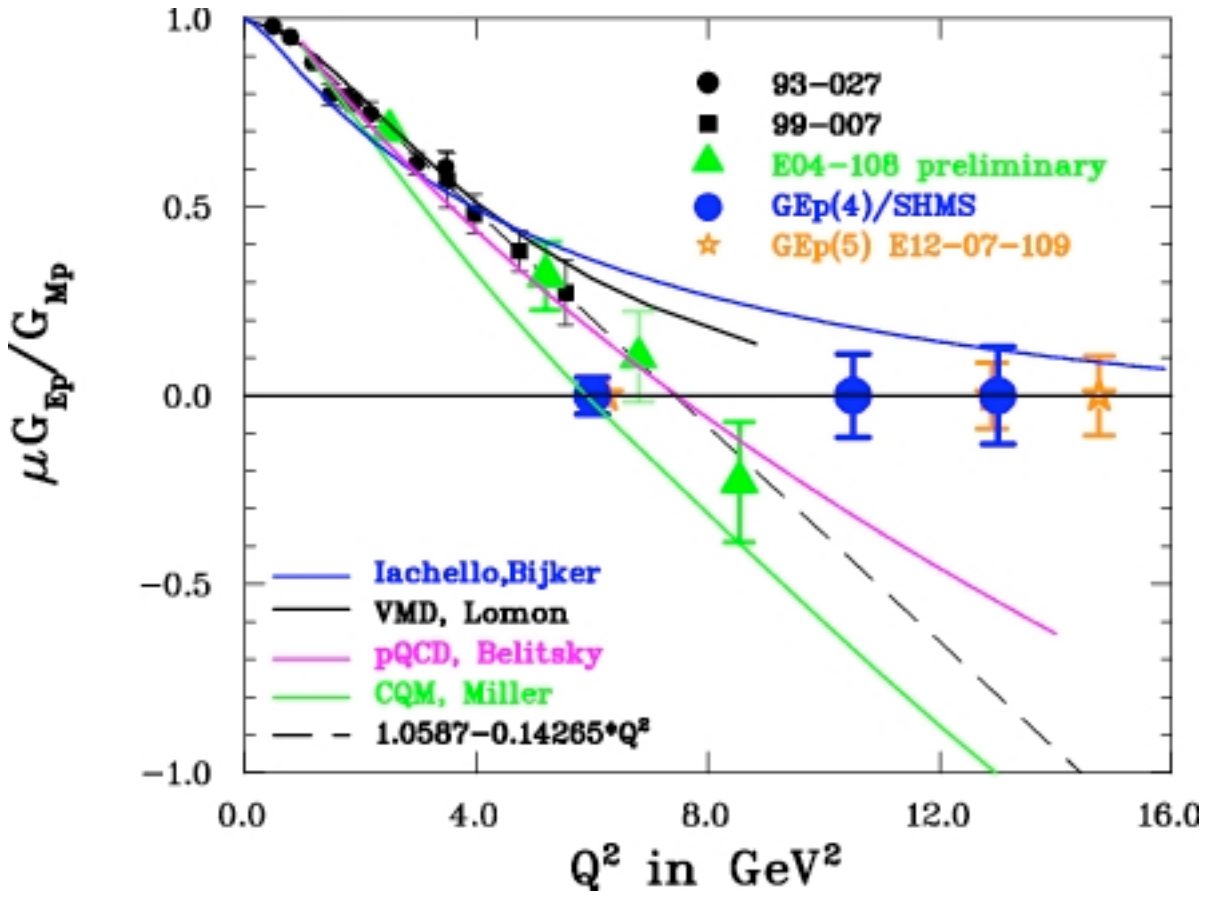
Gep-IV: Kinematics

Q^2	E_e	θ_e	$E_{e'}$	θ_p	p_p	$d\sigma/d\Omega_e$	ϵ	χ	$\Delta\Omega_e$
GeV^2	GeV	deg	GeV	deg	GeV/c	cm^2/sr		deg	msr
6	6.6	30	3.4	25	4.03	1.1×10^{-35}	0.72	145.4	8.6
10.5	8.8	35.5	3.20	16.7	6.47	3.5×10^{-37}	0.55	229.7	24
13	11.0	31.3	4.07	15.7	7.81	1.6×10^{-37}	0.58	276.5	23

Table 2: *The proposed kinematics. Assumed SHMS spectrometer solid angle: 5 msr. Assumed beam characteristics: 75 μA, 85% polarization. Assumed target: 30 cm LH_2.*

Q^2	E_e	COM	absolute $\Delta(G_{Ep}/G_{Mp})^*$	time
GeV^2	GeV			days
6.0	**6.6**	**3.9×10^{-3}**	**0.04**	**4**
10.5	**8.8**	**1.5×10^{-3}**	**0.11**	**30**
13.0	**11.0**	**1.1×10^{-3}**	**0.13**	**60**

Gep-IV: Predictions



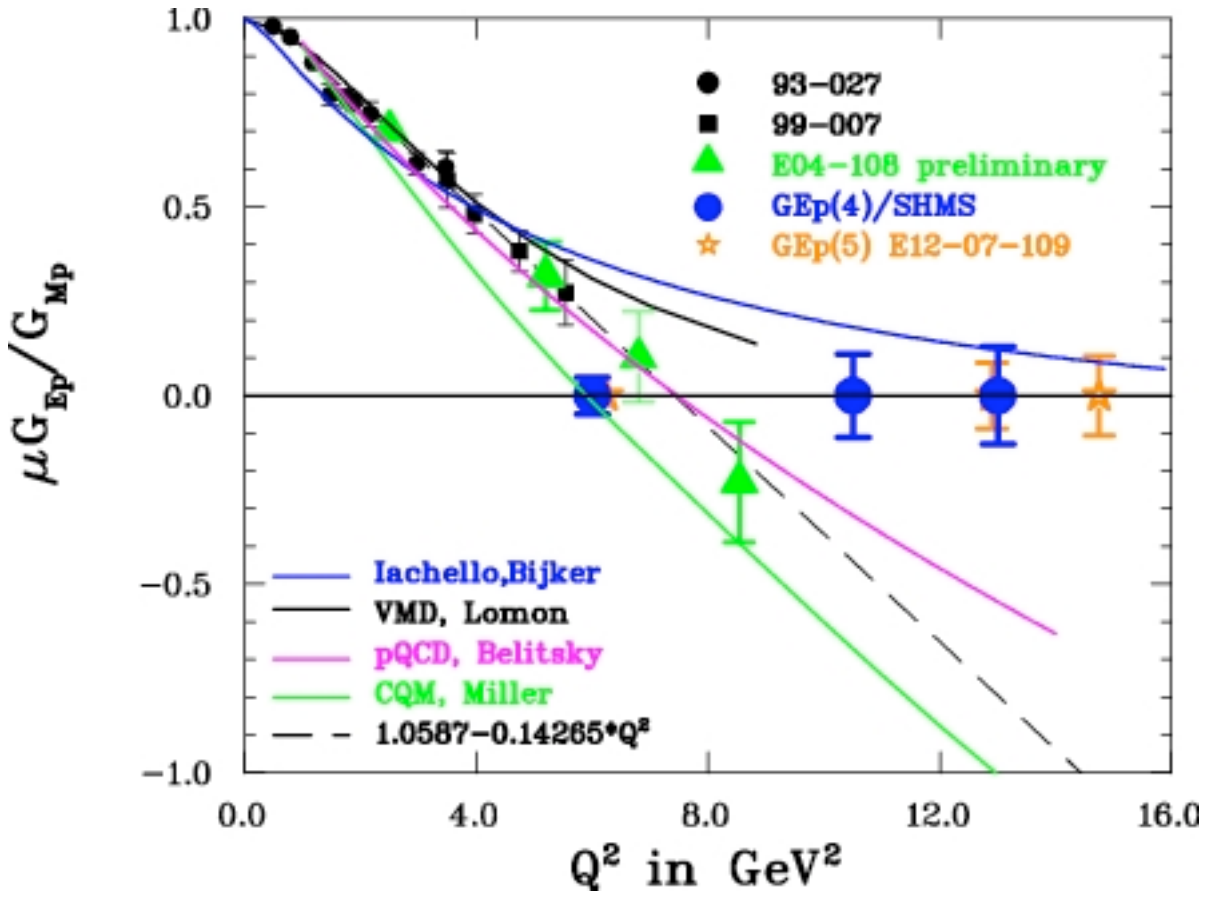
Gep-IV vs. Gep-V

- Clean identification of elastic events is of crucial importance
  - SHMS has better momentum resolution
  - SBS has better angular resolution and y_target resolution
- As a result, elastic identification will be accomplished in very different ways in the two experiments, resulting in different systematics
- SHMS will have much lower level of random/accidental background
- Gep-IV uses existing/base Hall C equipment - this has technical as well as financial/funding advantages
- Gep-IV can be run very early post-upgrade
- Software algorithms are refined and well-tested

Conclusions

- Many competing/complementary theoretical models, with different approaches
- While most modern calculations describe the data well in the lower Q^2 regime, they begin to diverge significantly beyond the currently available data
- New data at higher Q^2 (for both proton and neutron) will place stringent constraints on available models, and will continue to motivate more advanced calculations
- Gep-IV will provide high quality data on the form factor ratio up to Q^2 = 13 GeV^2, using existing/base equipment. The experiment can be carried out as soon as the 11 GeV electron beam is available in Hall C - could even be used as a commissioning experiment for the SHMS
- No major technical issues face this experiment - hardware and software are "ready to go"

Gep-IV: Predictions



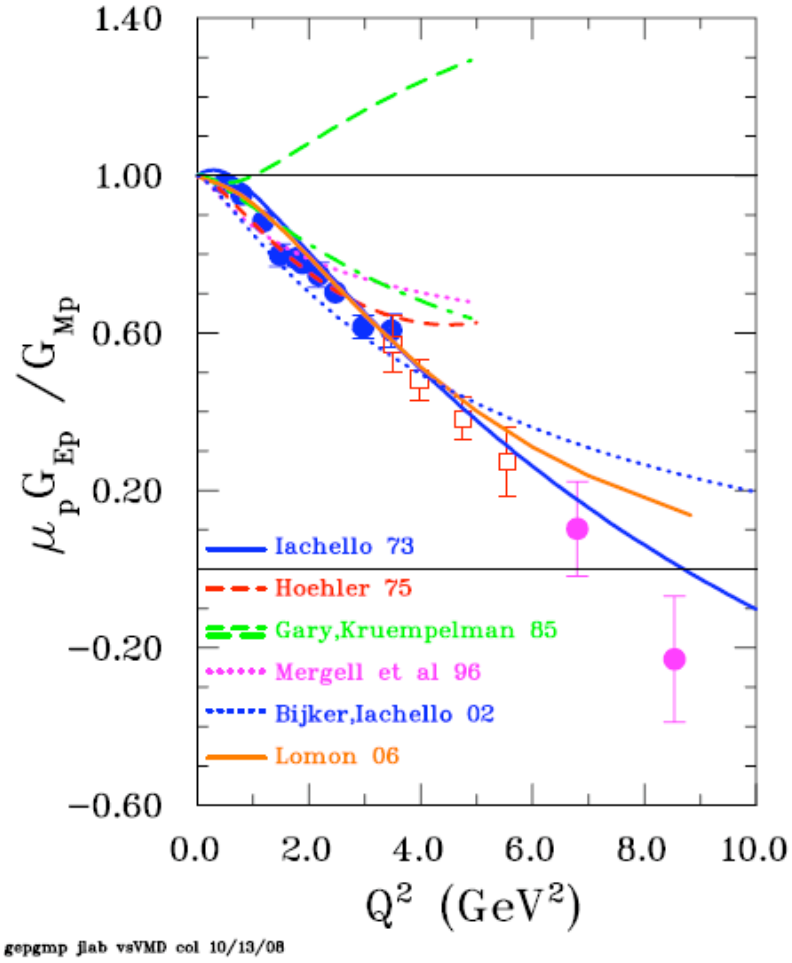
Backup Slides

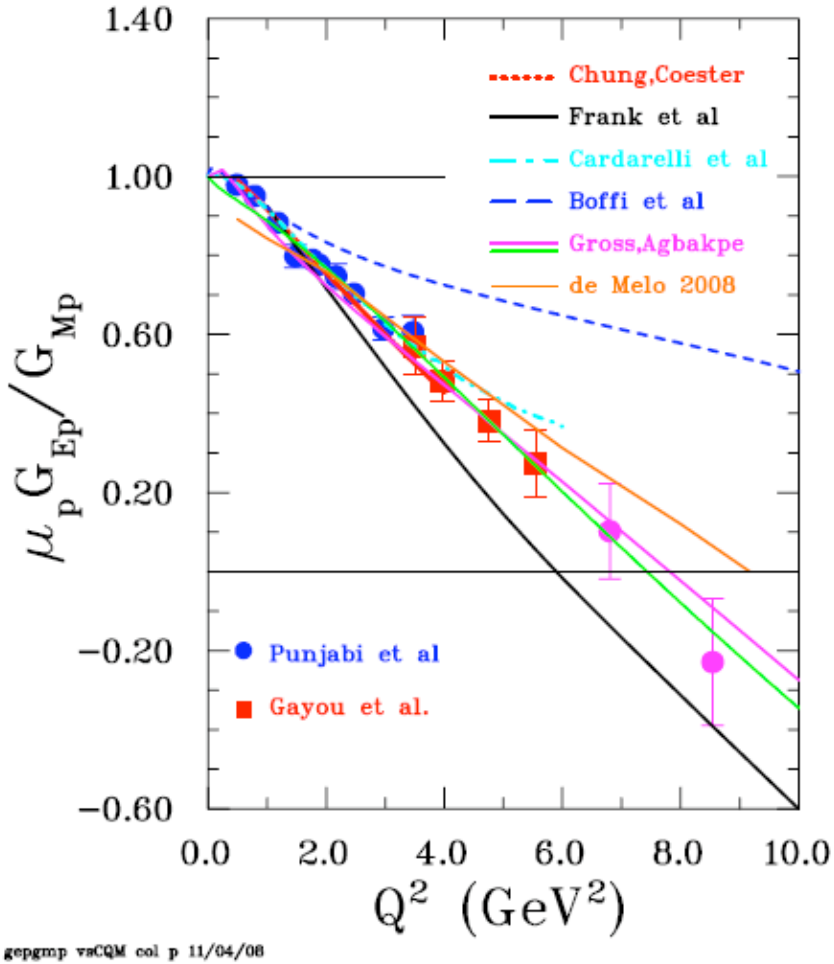
Q^2	E_e	COM	Horiz. Bender	$\Delta(G_{Ep}/G_{Mp})^*$	time
GeV^2	GeV		Factor		days
6.0	**6.6**	**3.9×10^{-3}**	**1.15**	**0.04**	**4**
9.0	8.8	2.0×10^{-3}	1.70	0.11	30
10.5	**8.8**	**1.5×10^{-3}**	**1.30**	**0.11**	**30**
12.0	8.8	1.2×10^{-3}	1.01	0.12	60
12.0	11.0	1.2×10^{-3}	1.01	0.12	60
13.0	**11.0**	**1.1×10^{-3}**	**1.00**	**0.13**	**60**
14.0	8.8	0.9×10^{-3}	1.06	0.20	120
14.0	11.0	0.9×10^{-3}	1.06	0.16	120

Table 6: *Absolute uncertainties (not including systematics), and times required. The assumed beam intensity and electron beam polarization are 75 μA and 0.85, respectively. The target length is 30 cm, and the SHMS solid angle is 5.0 msr.*

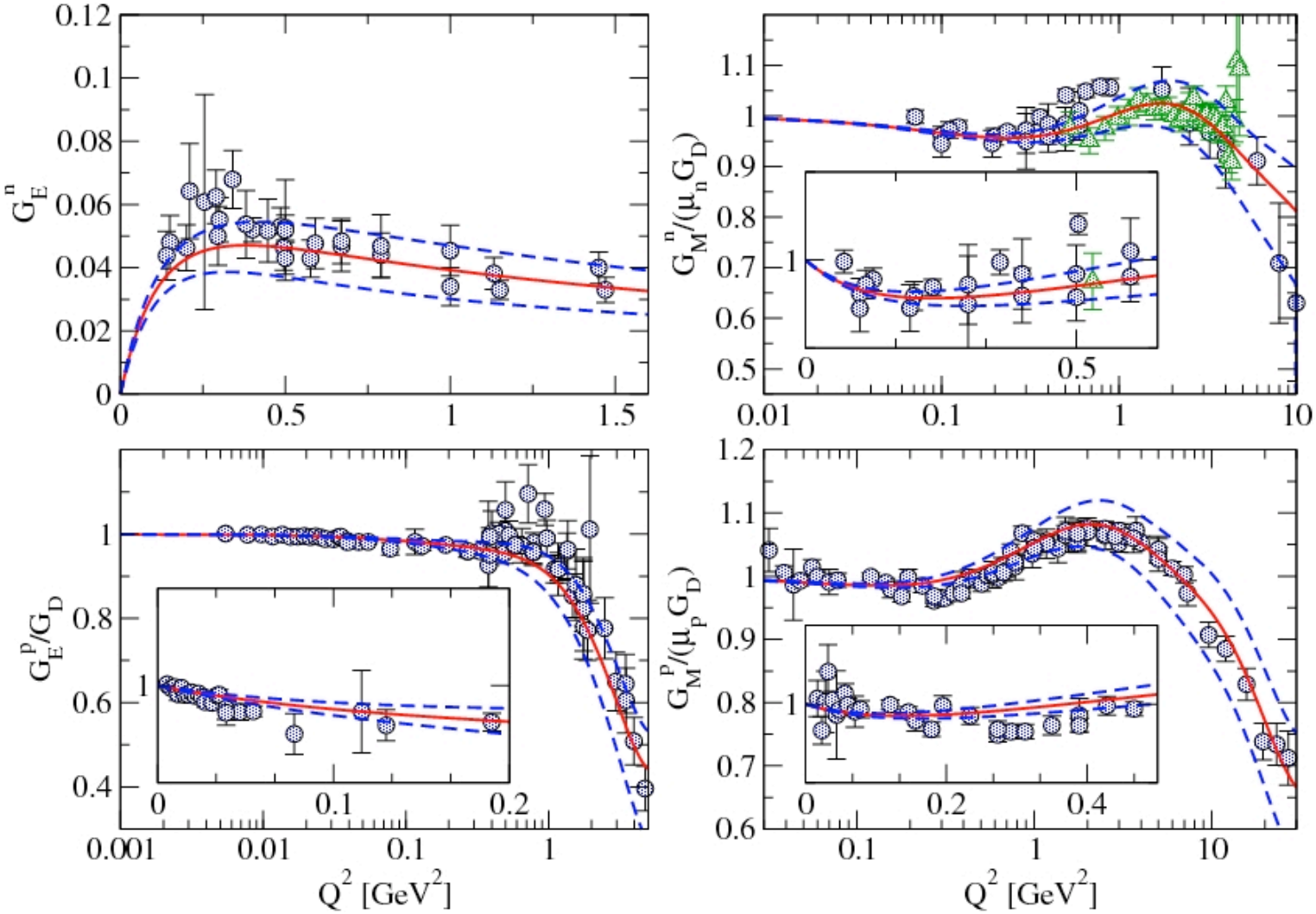
* *Note that the increase in the error bar due to precession in the horizontal bender has been included.*

VMD and CQM


1.40
1.00
0.60
0.20
−0.20
−0.60
0.0
2.0
4.0
6.0
8.0
10.0
$\mu_p G_{Ep} / G_{Mp}$
Q^2 (GeV^2)
Iachello 73
Hoehler 75
Gary,Kruempelman 85
Mergell et al 96
Bijker,Iachello 02
Lomon 06
gepgmp jlab vsVMD col 10/13/08


1.40
1.00
0.60
0.20
−0.20
−0.60
0.0
2.0
4.0
6.0
8.0
10.0
$\mu_p G_{Ep} / G_{Mp}$
Q^2 (GeV^2)
Chung,Coester
Frank et al
Cardarelli et al
Boffi et al
Gross,Agbakpe
de Melo 2008
Punjabi et al
Gayou et al.
gepgmp vsCQM col p 11/04/08

Dispersion Relation Fits (Belushkin *et al.*)

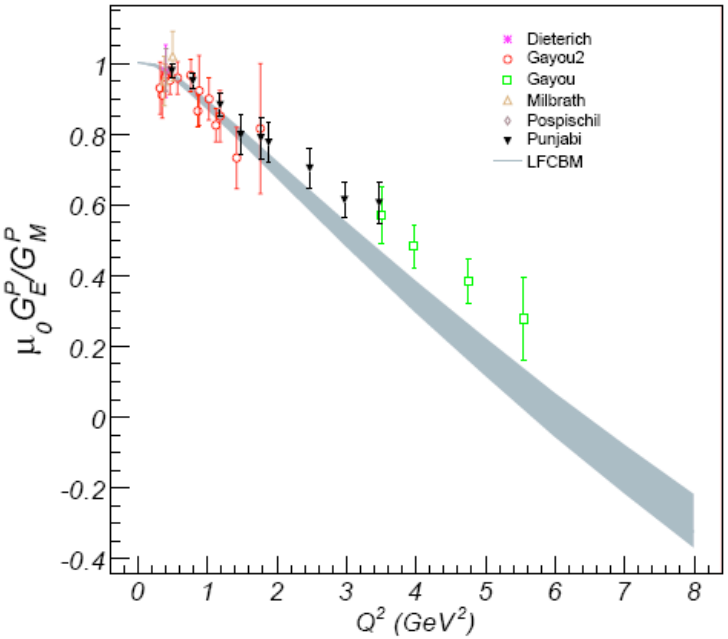


Lattice QCD, Fadeev

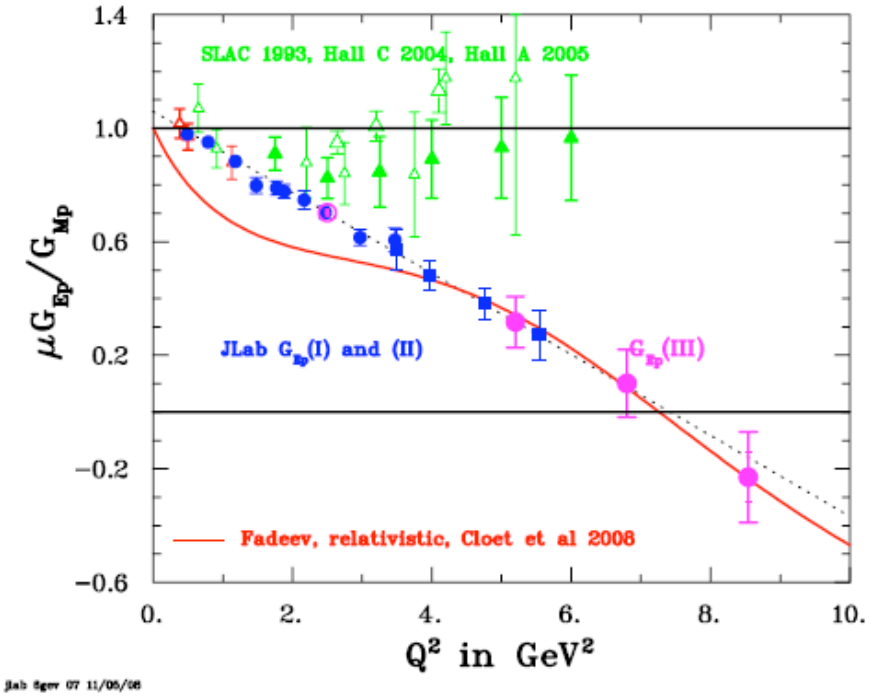
For example:

Matevosyan, Thomas,Miller(AIP 2005)
Use LFCBM (Light Front Cloudy Bag model, rCQM) to extrapolate QCDSF 'data' to physical pion mass, (Gloeckeler et al 2005)

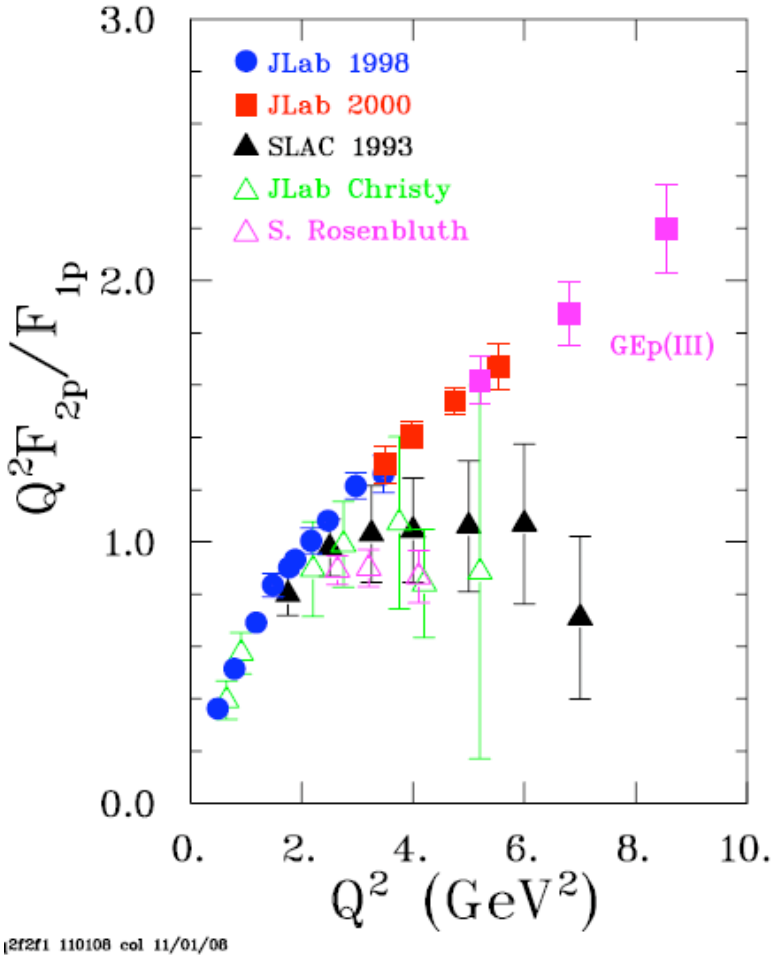
Poincare invariant Faddeev equation solution.
Arrington,C. Roberts and Zanetti et al.
J. Phys. G34, S23 (2007)
Cloet et al t b p 2008
Parameters: diquark magnetic dipole and quadrupole moments, scalar to vector EM transition strenath.

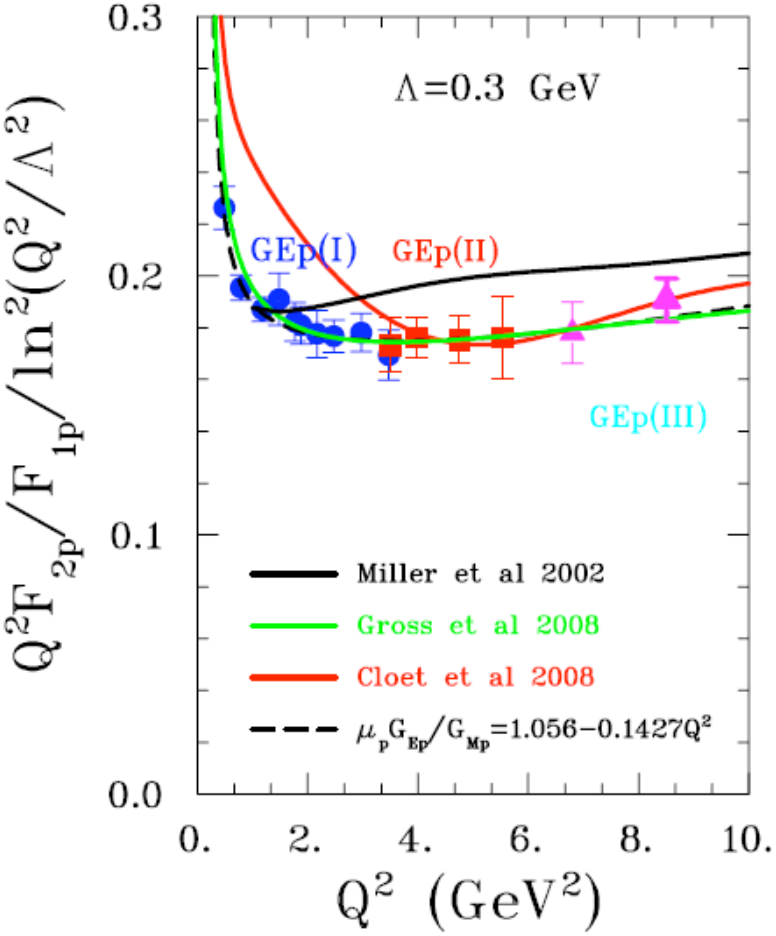


H.H. Matevosyan - PANIC 2005



Proton: F2 /F1 and pQCD

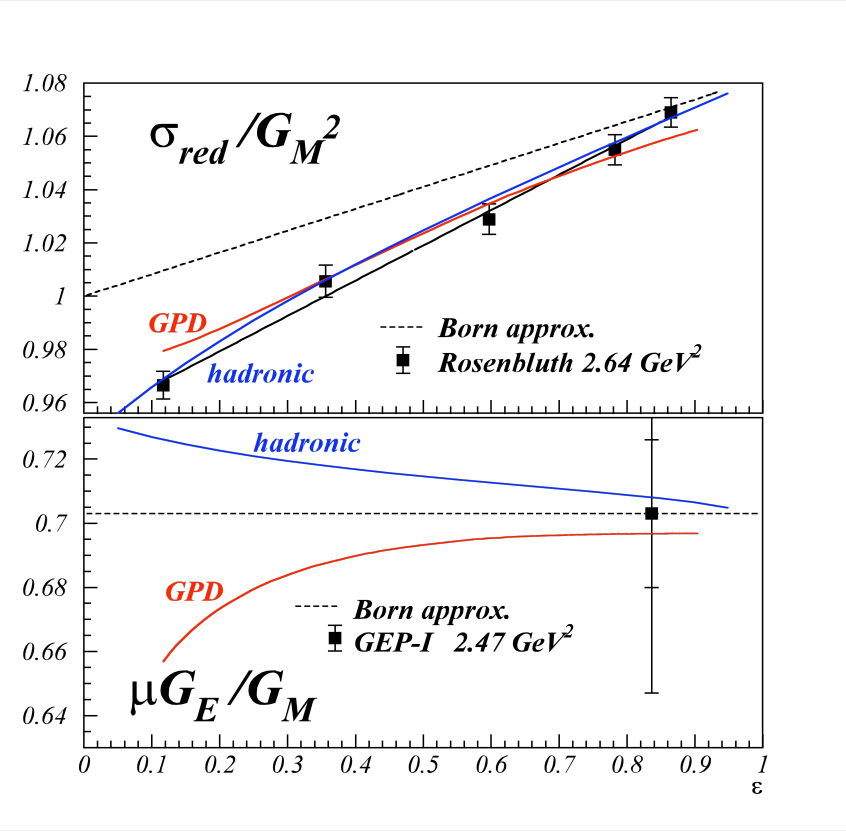


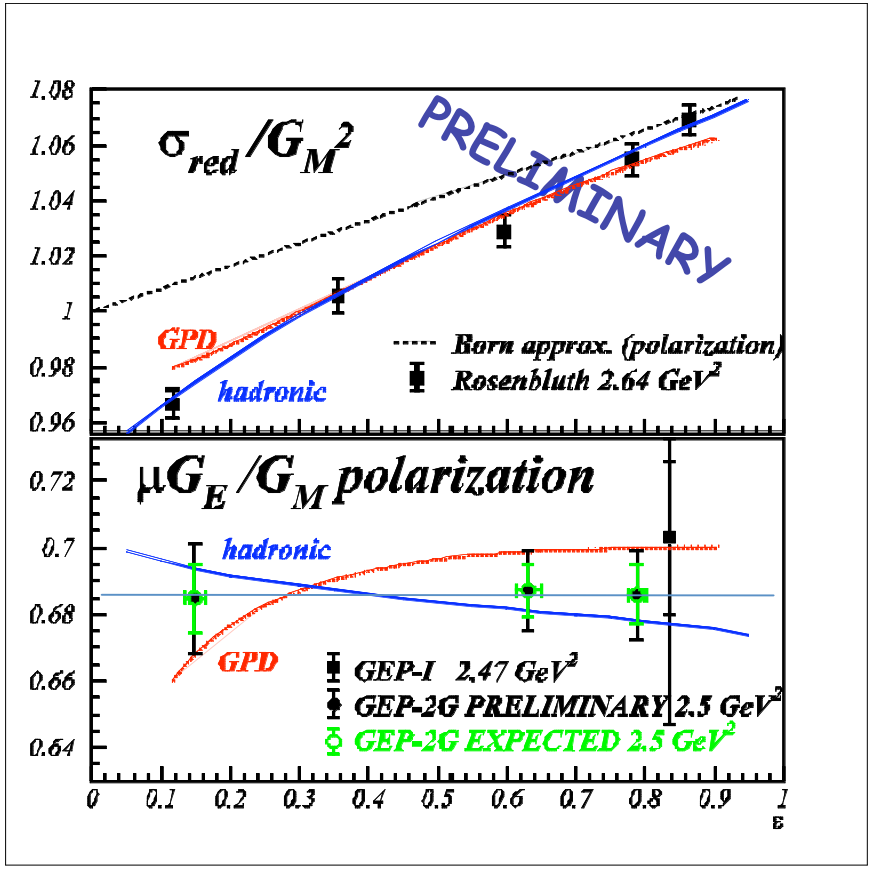


Brodsky and Farrar (75):
$Q^2F_2/F_1 \rightarrow$ constant

Belitsky, Ji and Yuan (03):
$Q^2F_2/F_1 \rightarrow \ln^2(Q^2/\Lambda^2)$

2 γ -Gamma Models and Data

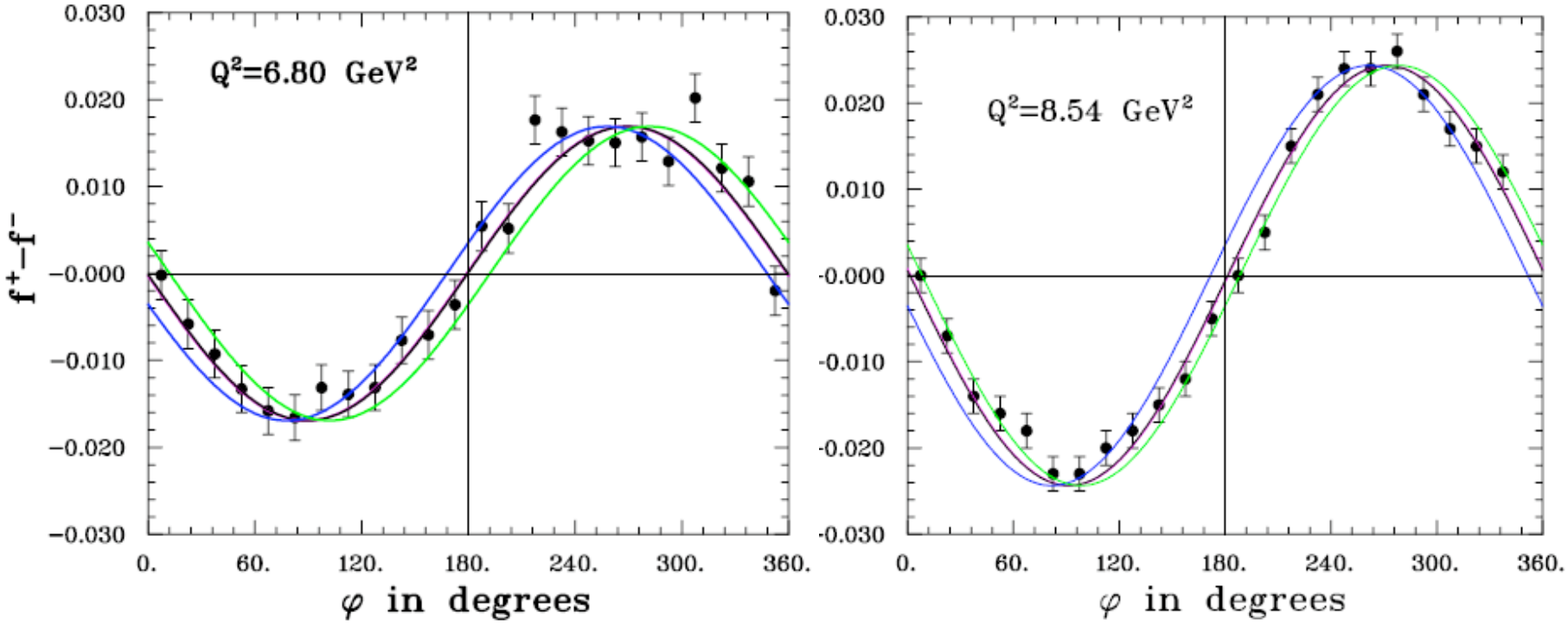




The inclusion of hard 2 γ exchange:
Chen et al (2003) with GPDs,
Blunden et al (2003) in hadronic model
Several other calculations exist!
Predictions range between no effect and approximative reproduction of data.

Expt. Jlab 04-019 measured Ratio at 3 values of ε for Q^2=2.49 GeV^2; no ε-dependence at 0.01 level

Physical Asymmetries at Q^2 of 6.8 and 8.5 GeV2



Constituent Quark Models (Understanding the Dynamics)

Initially proposed by Isgur & Karl (78)
Early non-relativistic CQM

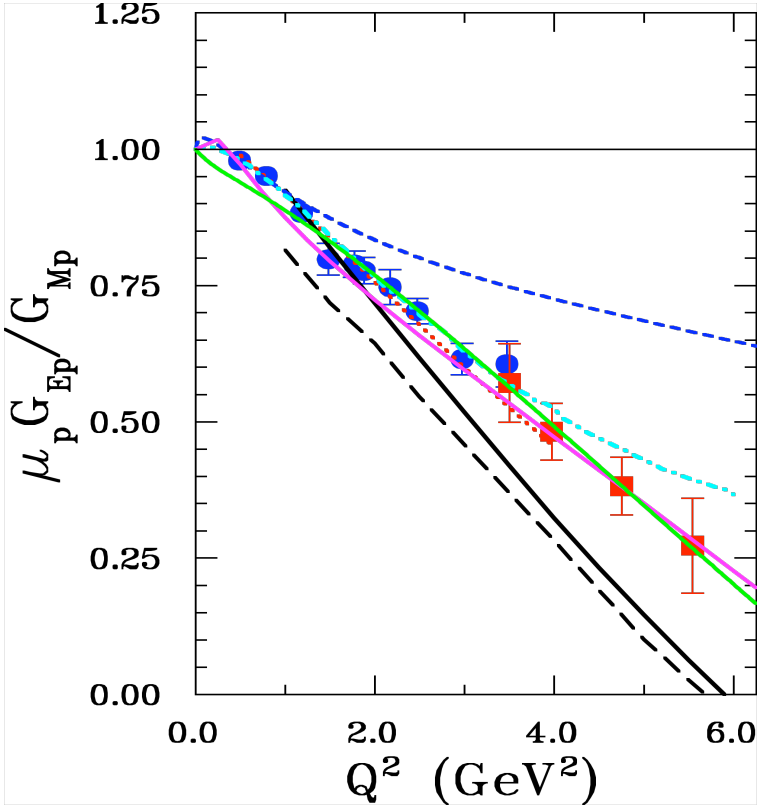
Relativistic CQM: 3 forms of dynamics: instant, point like and light-front

Point-like: Chung & Coester (91) .....
Cardarelli et al (00) ——

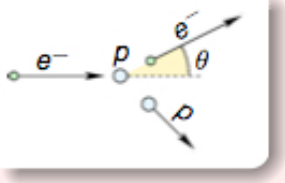
Front form:
Frank,Jennings, Miller (96) – – – –
G.A. Miller, Frank (02) ——

Goldstone boson: Boffi et al (01) – · –

Cov. spectator: Gross,Agbakpe (04) – – –
Gross,Ramalho,Pena (06) ——

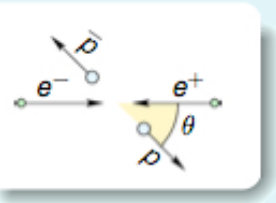


Conclusion: Relativistic CQ are relevant degrees of freedom



Elastic scattering

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 E_e' \cos^2\frac{\theta}{2}}{4E_e^3 \sin^4\frac{\theta}{2}} \left[G_E^2 + \tau \left(1 + 2(1+\tau)\tan^2\frac{\theta}{2} \right) G_M^2 \right] \frac{1}{1+\tau}$$

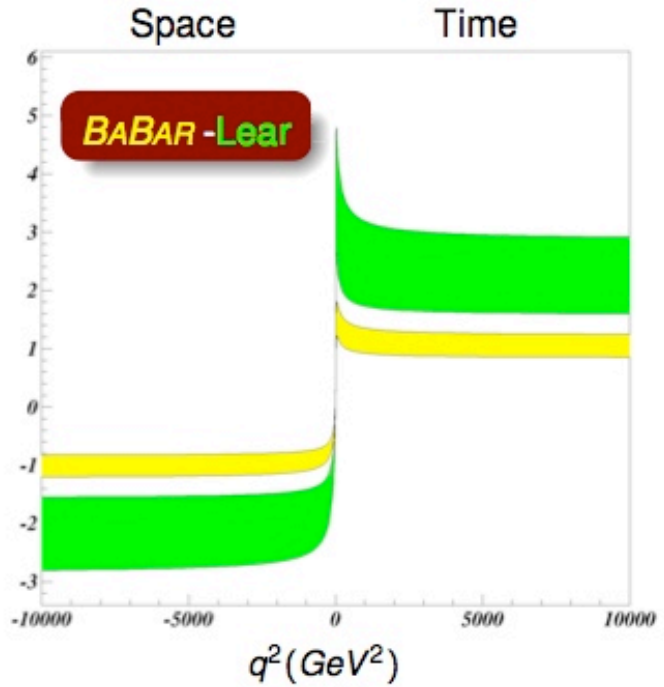


Annihilation

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 \sqrt{1-1/\tau}}{4q^2} C \left[(1+\cos^2\theta)|G_M|^2 + \frac{1}{\tau}\sin^2\theta |G_E|^2 \right]$$

Coulomb correction: $C \approx \frac{y}{1-e^y}$ $\qquad y = \frac{\pi\alpha M_p}{\beta q}$

Asymptotic behaviour of $G_E^p(q^2)/G_M^p(q^2)$



pQCD prediction

$$\left| \frac{G_E^p(q^2)}{G_M^p(q^2)} \right| \xrightarrow[|q^2|\to\infty]{} 1$$

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